

Dynamic Emergency Vehicle Prioritization in a Connected Urban Environment: A Rule-Based Approach

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Abstract. Efficient and rapid management of Emergency Vehicles (EVs) such as ambulances, fire trucks, and police vehicles is crucial for saving lives and protecting property. Road intersections often become bottlenecks, hindering the swift movement of EVs despite their priority. This paper proposes an Emergency Vehicle Priority Management (EVP) system leveraging IoV, 5G connectivity, GPS, and V2X communication. The system employs a rule-based algorithm and real-time data to dynamically adjust traffic signals and vehicle lanes, minimizing disruptions to general traffic while prioritizing EVs. Simulations conducted using the SUMO traffic simulator and real-world traffic data from TAPAS Cologne demonstrate significant improvements in travel times, waiting times, and overall traffic flow efficiency. Specifically, the EVP system achieved reductions in average travel times by 367s, waiting times by 31s/km, waiting counts by 9, and time loss by 40s/km, which correspond to an improvement of 27.18%, 60.91%, 59.42%, and 53.54% respectively. These results highlight the potential of the proposed system to enhance urban traffic management and emergency response efficiency.

Keywords. 5G, Emergency Vehicle Priority, IoV, SUMO, Traffic Management, V2X

1. Introduction

Efficient traffic management for emergency vehicles (EVs) like ambulances, fire trucks, and police vehicles is crucial for timely incident responses, essential for saving lives and protecting property. Timely incident response is paramount, yet road intersections often become bottlenecks, hindering the swift movement of EVs despite their priority status. This results in significant delays and inefficiencies in emergency response, potentially exacerbating injuries, increasing property damage, and causing significant economic losses. Traditional traffic management systems often struggle to effectively prioritize EVs, relying on static approaches that are often insufficient during peak traffic hours or when unexpected situations arise. This highlights the urgent need for more advanced

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and adaptive solutions that can dynamically adjust to changing traffic conditions.

Recent advancements in communication technologies, particularly the Internet of Vehicles (IoV) and 5G connectivity, offer promising avenues to address these challenges. IoV facilitates seamless communication between vehicles and infrastructure, while 5G technology provides the high-speed, low-latency data transmission required for real-time information exchange. This paper proposes an innovative Emergency Vehicle Priority Management (EVP) system that leverages a rule-based approach within a hybrid architecture integrating IoV and 5G technologies. The system focuses on two critical aspects: dynamic traffic signal management and lane clearance, ensuring that EVs can navigate intersections swiftly and safely while minimizing disruption to general traffic flow. In recent years, significant research has been conducted in the field of traffic management systems aimed at prioritizing emergency vehicles. In [1], authors introduced an IoT-based intelligent traffic management system that prioritizes vehicles based on incident type and addresses traffic signal hacking. This system demonstrated superior performance in congestion avoidance and travel time reduction, meeting the response time targets set by the Department of Treasury and Finances Budget and Financial Management Guidance (BFMG). Rerearch work in [2] presents an IoT-based intelligent traffic signal system that manages traffic signals through devices like Raspberry Pi and Node MCU, effectively reducing waiting times despite the high infrastructure costs involved. In another work, authors focused on reducing emergency vehicle travel times using Dedicated Short-Range Communications (DSRC) in an emergency vehicle-centered traffic signal control system within intelligent transportation systems [3]. In [4], the presented work introduced a novel road traffic management strategy based on V2V communications aimed at reducing delays for emergency vehicles. It highlighted the potential of V2V communication for improving traffic flow, though there are some limitations in non-dedicated lanes. Moreover, in [5], the authors proposed a real-time traffic signal control strategy for emergency vehicles that leverages IoT and real-time communication to dynamically adapt traffic signals. This approach significantly improves response times and efficiently processes real-time data. In [6], a Real-Time V2I-based Emergency Vehicle Traffic Signals Management System (EVTMS) was presented. This system provides free travel paths for emergency vehicles but requires substantial V2I infrastructure for effective deployment. In [7], researchers developed an emergency vehicle recognition system using YOLOv8 for object detection, achieving real-time performance but facing computational intensity issues. The work presented in [8] is a development of the EVATL framework, integrating adaptive traffic lights with emergency vehicle communication using a combination of IoT, GPS, and YOLOv8 for real-time detection and communication, demonstrating significant improvements in waiting times. Furthermore, in [9], the authors discussed an adaptive traffic signal control system for emergency vehicle prioritization using IoT technology. This system aims to reduce delays and pollution in metropolitan cities by dynamically adjusting traffic signals, though the complexity of implementation posed a significant challenge. A smart traffic management system using multi-sensor integration for congestion control and emergency vehicle priority was proposed in [10]. This enhanced ambulance control and improved response times in urban environments despite challenges in sensor integration. Finally, in [11], the authors explored traffic light coordination and auctions for intersection management in smart city scenarios. Their approach, tested on both synthetic and real urban maps in Modena, Italy, outperformed classical

Fixed Time Control systems and auction strategies that did not exploit traffic light coordination.

While significant research has been conducted in the field of traffic management systems aimed at prioritizing emergency vehicles, existing methodologies often lack the flexibility and adaptability required for optimal EV prioritization. Existing systems, like those cited above, often focus on either traffic signal optimization or lane prioritization, but not both dynamically. Our proposed EVPM system addresses these gaps by dynamically adjusting traffic signals and lane management in real-time, using a combination of IoV, 5G connectivity, GPS, and RFID. This dynamic and responsive approach contrasts with static or less adaptive methods seen in previous works. Moreover, our simulation approach leverages the SUMO traffic simulator and real-world data from TAPASCologne to create a realistic urban traffic environment, providing a comprehensive evaluation of the system's effectiveness. By incorporating a wide range of traffic conditions and emergency scenarios, this methodology allows for a more accurate assessment of the system's impact on both EV response times and general traffic flow.

The structure of this paper is as follows: Section 2 outlines the Proposed Methodology, focusing on the architecture and algorithm of our EVPM system. Section 3 presents the Simulation Setup and Procedure. Section 4 presents the Results and Discussion, evaluating the performance of our method through simulation and comparing it with existing approaches. Finally, Section 5 concludes with a discussion on the effectiveness of our proposed system and outlines future research directions.

2. Proposed Methodology

The proposed method for intelligent traffic management for emergency vehicles (EVs) leverages a rule-based approach and a hybrid architecture that integrates 5G connectivity and the Internet of Vehicles (IoV). The system assumes a high penetration of connected vehicles (CVs), with each road equipped with a control unit known as a Road Side Unit (RSU). The RSUs receive real-time data from vehicles and traffic lights to identify EVs and execute a rule-based algorithm to facilitate their passage through intersections. The system addresses two critical aspects: traffic signal management and lane clearance, ensuring that EVs can navigate through intersections quickly and safely while minimizing disruption to general traffic.

2.1. Network Architecture

The proposed network architecture aims to enhance communication and coordination between vehicles and infrastructure through the Internet of Vehicles (IoV). Each vehicle is equipped with a GPS module and an On-Board Unit (OBU) that communicate with Roadside Units (RSUs) using 5G-V2X (Vehicle-to-Everything) technology. The OBU collects critical data such as vehicle speed, ID, GPS coordinates, and lane index, transmitting this information to the RSUs in real-time.

To improve vehicle detection and identification, the system integrates various sensors, including RFID, ultrasonic, and infrared sensors, along with GPS data from the ve-

hicles. RFID sensors capture the unique RFID tags of vehicles, providing precise identity information. Ultrasonic and infrared sensors, installed in the vehicles, determine the exact lane positions. This comprehensive data collection ensures accurate identification and monitoring of vehicles within the RSU-controlled zone.

The RSUs process this integrated data to manage traffic lights and lane assignments effectively, ensuring a smooth flow of traffic, particularly for emergency vehicles. The high-speed, low-latency communication facilitated by 5G-V2X technology is crucial for the timely and reliable exchange of information necessary for real-time traffic management. For communication with traffic lights, RSUs may relay through the gNodeB (gNB) of the 5G network if the traffic light is beyond the direct range of the RSU. Communications between vehicles and RSUs are categorized as V2I (Vehicle-to-Infrastructure), between RSUs and gNB as I2N (Infrastructure-to-Network), and between gNB and traffic lights as N2I (Network-to-Infrastructure) (Figure 1).

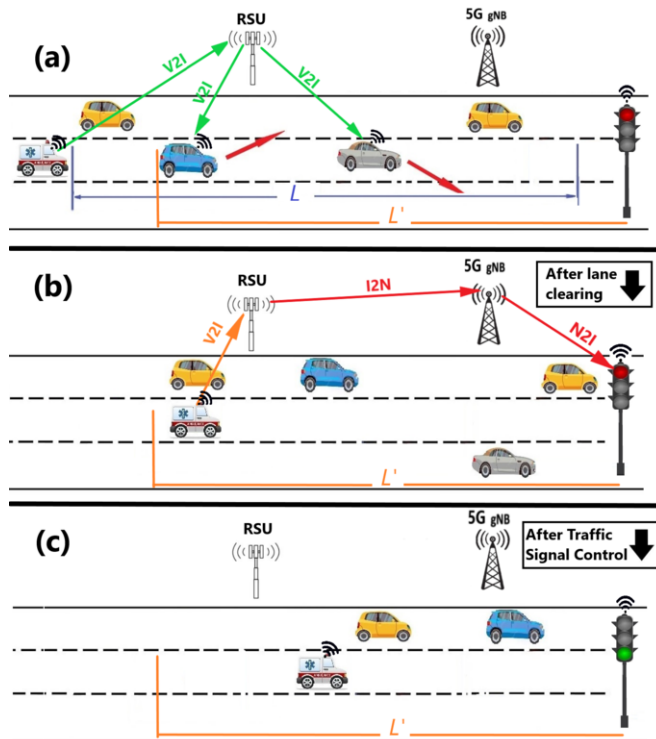


Figure 1. Real-Time Emergency Vehicle Management using 5G and V2X Communication.

2.2. Emergency Vehicle Detection and Tracking

Upon the entry of a vehicle into the RSU-controlled zone, several detection and tracking steps are implemented to ensure the efficient passage of emergency vehicles:

RFID Detection: As soon as a vehicle enters the zone, RFID readers installed along the roads capture the signal from the vehicle's unique RFID tag and transmit its identifier to the RSU.

Identity Validation: The RSU uses this identifier to determine whether the vehicle is an emergency vehicle, helping differentiate EVs from regular vehicles.

Real-Time Tracking: Once identified, the RSU continuously tracks the position and lane index of the emergency vehicle using data from ultrasonic and infrared sensors, as well as GPS data from the OBUs. This continuous monitoring is essential to determine the necessary actions to facilitate the emergency vehicle's movement.

Data Reception and Processing: The RSUs continuously receive and process data transmitted by the OBUs in the vehicles. These OBUs collect information from various sensors, including ultrasonic and infrared sensors, along with GPS data. The RSUs use this data to monitor all vehicles on the same road as the emergency vehicle. For each detected vehicle, the RSU gathers information on its ID, position, and lane index.

Lane Clearance and Traffic Signal Control: If a vehicle is in the same lane as an emergency vehicle and within a distance less than the Lane Clearance Distance (L), the RSU sends a command for the vehicle to change lanes, thereby clearing the way for the emergency vehicle (Figure 1-b). In scenarios of severe congestion, the lane control may not function optimally. However, it is highly effective in most cases. Even if vehicles cannot yield immediately, the system remains engaged and clears the lane once possible. This ensures that emergency vehicles are prioritized as soon as conditions permit.

Additionally, if the road is controlled by a traffic light and the emergency vehicle is within the Traffic Light Detection Zone (L'), set to 100 meters, the RSU changes the light to green to allow the emergency vehicle to pass without delay (Figure 1-c). This distance was estimated empirically after several simulations to balance the need for emergency vehicle priority and the minimization of disruption to general traffic. Beyond 100 meters, frequent changes in traffic light signals could cause significant disruptions to the overall traffic flow.

When the light turns green for the lane with the emergency vehicle, the other branches of the intersection will be temporarily blocked. Although this may cause delays for the other lanes, prioritizing emergency vehicles is crucial and justified, as their passage is rare and essential for rapid response in critical situations. The overall impact on general traffic remains limited.

For the purposes of this study, the following distances are defined:

- **Lane Clearance Distance (L):** The distance ahead of an emergency vehicle within which other vehicles in the same lane must change lanes to clear the way. This distance is calculated based on the braking distance formula, which takes into account the speed (v) of the emergency vehicle:

$$L = \frac{v^2}{2 \cdot \mu \cdot g} \quad (1)$$

where μ is the coefficient of friction, typically taken as 0.7 for dry asphalt, and g is the acceleration due to gravity (approximately 9.81 m/s^2) [12].

- **Traffic Light Detection Zone (L'):** A fixed distance of 100 meters from a traffic light within which the signal state change is tolerated to allow the emergency vehicle to pass without delay.

The effectiveness of the EVPM system is illustrated in the flowchart and algorithm (Figures 1 and 2).

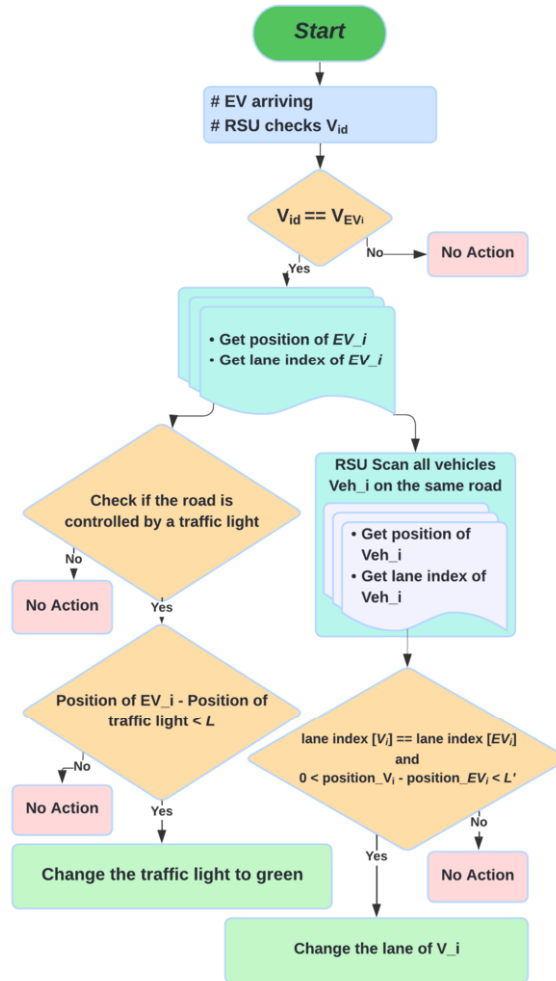


Figure 2. Workflow of the EVPM system

3. Simulation Setup and Procedure

3.1. Simulation Environment

To validate the effectiveness of the proposed system, we conducted a series of simulations using the SUMO traffic simulator. SUMO is an open-source, highly portable, microscopic, and continuous road traffic simulation package designed to handle large road networks. We used it to create a realistic urban traffic environment, specifically modeling the city of Cologne, Germany. The simulation utilized the TAPASCologne scenario, which provides detailed traffic data for the city. This dataset includes 24 hours of traffic data, representing daily traffic in the entire city, with a total of 1,549,611 vehicles. The simulated network comprehensively models Cologne's road infrastructure, including multiple intersections and traffic lights. TAPASCologne was chosen for its comprehen-

Algorithm 1 EVPM's algorithm

Input: State of all vehicles
Output: Adjusted states of vehicles and traffic lights TL for emergency vehicle priority

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1: procedure EVPM
2:   # Initialize the system
3:   # Detect emergency vehicle arrival
4:   for  $V_i$  on the RSU-controlled zone do
5:     # RSU checks  $V_{id}$ 
6:     if  $V_{Id} == EV_{id}$  then
7:       Get position of  $EV_i$ 
8:       Get lane index of  $EV_i$  :  $EV_i$  L.ind
9:       for  $V_i$  on the same road do
10:        Get position of  $V_i$ 
11:        Get lane index of  $V_i$  :  $V_i$  L.ind
12:        if  $V_i$  L.ind ==  $EV_i$  L.ind then
13:          if pos  $V_i$  - pos  $EV_i$  <  $L$  then
14:            Change the lane of  $V_i$ 
15:          end if
16:        end if
17:      end for
18:      if the road is controlled by a TL then
19:        if pos  $EV_i$  - pos TL <  $L'$  then
20:          Change the traffic light to green
21:        end if
22:      end if
23:    end if
24:  end for
25: end procedure

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sive and realistic representation of urban traffic patterns, making it highly representative of typical urban traffic conditions and ensuring the validity of the simulation results.

3.2. Traffic Flow and Vehicle Distribution

The traffic data was divided into hourly segments, and simulations were conducted for each hour separately. For example, separate simulations were run for the periods from 6:00 to 7:00, 7:00 to 8:00, and so on, up to 18:00 to 19:00. Each hourly simulation included all vehicles from the TAPASCologne dataset for that specific hour. Additionally, 200 emergency vehicles were introduced into each hourly simulation. These emergency vehicles had different routes and departure times within the simulation hour, but their routes and departure times were kept consistent across all hourly simulations, with a 1-hour offset for each subsequent simulation.

3.3. Emergency Vehicle Characteristics

The simulation included various types of emergency vehicles, such as ambulances, fire trucks, and police cars. Emergency vehicles were given higher priority levels compared

to regular vehicles.

3.4. Traffic Light Control

Three separate simulations were conducted for each hour to evaluate different traffic light control strategies:

1. **Fixed-Time Scenario:** All traffic lights operated using fixed-time signal programs.
2. **Actuated Scenario:** All traffic lights operated using actuated signal programs.
3. **EVPM Scenario:** Traffic lights operated using actuated signal programs, except when an emergency vehicle (EV) was present, in which case the EVPM system was applied to prioritize the EV.

3.5. Data Collection

Data was collected on 200 EVs for several key metrics: travel time (in seconds), waiting time per kilometer (in seconds), waiting count which refers to the number of vehicles waiting at a certain point or on a particular road, and time loss per kilometer (in seconds). These metrics were gathered across three scenarios: fixed-time, actuated, and EVPM.

4. Results and Discussion

4.1. Simulation Results

4.1.1. Travel Time

Figure 3 shows the average travel time for vehicles from 6:00 to 18:00 across the three scenarios. The EVPM system significantly reduces travel time compared to both fixed-time and actuated signal programs. Notably, during the peak period of 7:00 to 8:00, travel time decreased from approximately 2144 seconds (fixed-time) and 1681 seconds (actuated) to 1411 seconds with the EVPM system.

4.1.2. Waiting Time

As depicted in Figure 4, the EVPM system consistently decreases the average waiting time per kilometer. The most significant reduction occurs between 7:00 to 8:00, with the average waiting time per kilometer dropping from 103 seconds (fixed-time) and 73 seconds (actuated) to 53 seconds with the EVPM system.

4.1.3. Waiting Count

Figure 5 compares the average waiting count for EVs across the three scenarios. The EVPM system effectively reduces the number of EVs waiting, particularly during the 7:00 to 8:00 period, where the count decreases from 34 (fixed-time) and 30 (actuated) to 19 with the EVPM system.

4.1.4. Time Loss

Figure 6 illustrates the average time loss per kilometer for EVs. The EVPM system shows a marked reduction in time loss, especially during the 16:00 to 17:00 period, where it decreases from 142 seconds (fixed-time) and 137 seconds (actuated) to 89 seconds with the EVPM system.

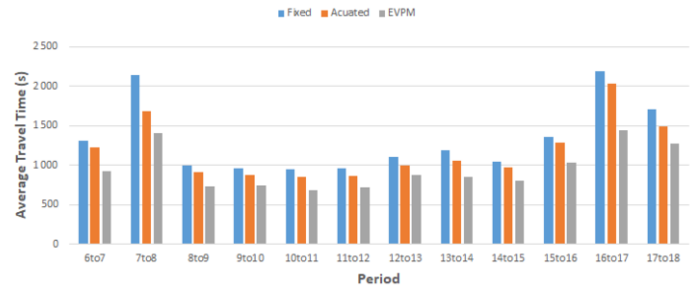


Figure 3. Comparison of travel time metric

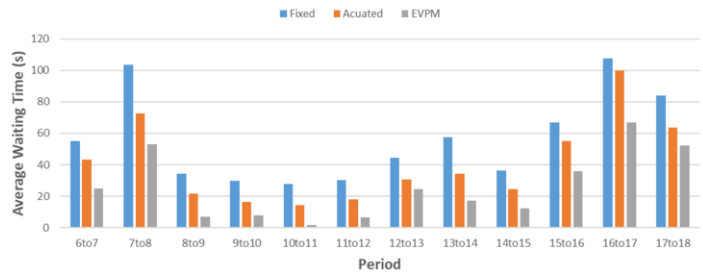


Figure 4. Comparison of waiting time metric

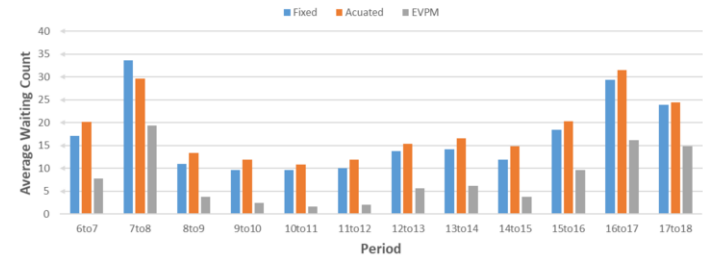


Figure 5. Comparison of waiting count metric

Table 1 provides the percentage improvements across the metrics, demonstrating the efficiency of the EVPM system in enhancing traffic flow and reducing delays.

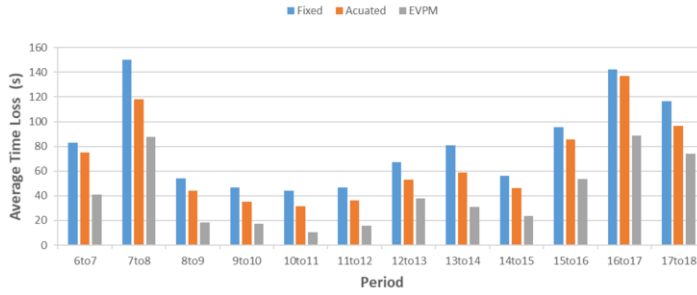


Figure 6. Comparison of time loss metric

Table 1. Improvements using EVPM

Metrics	Method	6-7	7-8	8-9	9-10	10-11	11-12	12-13	13-14	14-15	15-16	16-17	17-18	AVG
TT (%)	Fixed	0	0	0	0	0	0	0	0	0	0	0	0	0
	Aquated	6	22	8	9	10	10	11	11	7	5	10	13	10
	EVPM	29	34	26	22	28	25	29	29	33	23	25	27	27
WT (%)	Fixed	0	0	0	0	0	0	0	0	0	0	0	0	0
	Aquated	21	30	37	45	48	41	32	40	32	18	7	24	31
	EVPM	42	49	79	74	94	78	45	70	66	46	38	38	61
WC (%)	Fixed	0	0	0	0	0	0	0	0	0	0	0	0	0
	Aquated	-19	-12	-22	-24	-13	-20	-11	-17	-24	-10	-7	-2	-13
	EVPM	54	42	65	75	82	80	59	57	68	48	45	38	59
TL (%)	Fixed	0	0	0	0	0	0	0	0	0	0	0	0	0
	Aquated	9	21	19	25	29	23	21	27	18	11	4	17	19
	EVPM	51	42	66	63	76	66	47	62	58	44	38	36	54

4.2. Discussion

The results demonstrate the significant benefits of implementing the EVPM system in urban traffic management. The reduction in travel time, waiting time, waiting count, and time loss indicates that the EVPM system effectively prioritizes emergency vehicles, facilitating their swift passage through urban environments.

The most significant improvements are observed during peak traffic hours, such as 7:00 to 8:00 and 16:00 to 17:00, when traffic density is typically higher. During these times, the EVPM system's ability to dynamically manage traffic signals and reroute vehicles ensures that emergency vehicles can traverse intersections without delay, leading to faster response times and potentially reducing the risk of loss of lives and property damage.

The EVPM system's ability to reduce average travel time by approximately 27%, waiting time per kilometer by 61%, waiting count by 59%, and time loss per kilometer by 54% underscores its effectiveness in enhancing the passage of emergency vehicles, especially during periods of heavy traffic.

Overall, the EVPM system shows great potential for improving urban traffic management by ensuring the efficient passage of emergency vehicles. The significant improvements in key metrics, particularly during peak traffic hours, demonstrate its efficacy and highlight the importance of integrating advanced communication technologies in modern traffic management systems.

5. Conclusion

The proposed EVPM system represents a novel approach to urban traffic control. By seamlessly integrating IoT, GPS, and V2X communication, EVPM dynamically manages traffic signals and lane adjustments to prioritize emergency vehicles, resulting in substantial reductions in travel times, waiting counts, and time loss. This work makes several key contributions: the development of a novel EVPM system, the implementation of a comprehensive rule-based algorithm, and validation through extensive simulations utilizing real-world traffic data. While challenges such as high initial costs and data reliability remain, future research holds promise for further optimization. Investigating alternative decision-making algorithms, exploring deep reinforcement learning, and evaluating the system's performance in a wider range of traffic scenarios are promising areas for exploration. Furthermore, researching the feasibility of real-world deployment, conducting field trials to optimize parameters, and exploring integration with other intelligent transportation systems (ITS) components are crucial steps towards achieving more intelligent and efficient urban transportation systems. These future research directions aim to advance the field of emergency vehicle prioritization and contribute to the development of safer and more efficient urban environments.

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